

Simon Iradukunda

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EDUCATION

Amherst College

Expected, May 2029

Bachelor of Arts in Computer Science and Mathematics GPA: 3.79/4.00

Amherst, Massachusetts

- **Relevant Coursework:** Data Structures, Algorithms, OOP, Computer Systems, Applications of AI Engineering

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL, HTML/CSS

Technologies & Frameworks: FastAPI, PyTorch, React, HuggingFace Transformers, RAG, pandas, Playwright

Databases & Tools: PostgreSQL, Supabase, ChromaDB, SQLite, Docker, Git/GitHub, pytest, conda

EXPERIENCE

ASSISTments Foundation, Inc

June 2026 – Present

Software Engineer Intern, AI/ML

Worcester, MA, USA

- Shipped an AI feedback-image system to students on ASSISTments' national K–8 math platform by re-architecting a stalled generator into an SVG-native LLM pipeline.
- Built a multimodal RAG pipeline over the IM K–8 curriculum so feedback images matched teacher-guide strategy in **90%** of cases (LLM-judged), with grade-aware re-ranking in ChromaDB.
- Cut LLM evaluation costs **40%** while sustaining **85%** first-pass approval across **20K** problems by architecting an automated generate-validate-repair pipeline with deterministic safety guards.

Equity Bank Rwanda

Feb 2026 – May 2026

Software Engineer Intern

Remote(Kigali, Rwanda)

- Built an XGBoost model to predict loan approvals at **88%** accuracy, engineering predictive features from raw application data, correcting class imbalance, and systematically tuning hyperparameters.
- Reduced database query response time by 80% **from 25s to 5s** across internal tools by profiling execution bottlenecks, adding composite indexes, and rewriting complex SQL queries.
- Accelerated data retrieval **15%** for the bank's internal operations tools by optimizing REST API calls, and built responsive React components for the front end.

PROJECTS

Steam Semantic Search & Recommendation engine | Python, FastAPI, PostgreSQL/pgvector, Docker Dec 2025

- Delivered **<5ms** query latency across **50K+** games by architecting a hybrid async search backend fusing GIN full-text and HNSW vector indexes via SQL RRF.
- Resolved vector retrieval bottlenecks to hit **99.4%** recall@10 at **0.79ms** latency by engineering an automated parameter sweep against exact brute-force baselines.
- Ingested **41M+** user reviews into PostgreSQL by engineering an automated 4-stage ETL pipeline using Parquet columnar storage and idempotent asyncpg bulk upserts.

Deepfake & Fake-News Verifier | Python, PyTorch, HuggingFace, FastAPI

May 2026

- Achieved **<1s** latency across multi-modal RoBERTa + ViT by architecting an async, thread-pooled FastAPI service.
- Engineered a polymorphic ingestion pipeline parsing 5+ formats (docs, images, URLs), normalizing text+media into one scoring pipeline with **<50ms** overhead.
- Cut memory usage **60%** and eliminated cold-start latency by designing a singleton model registry with pre-warmed pipelines and SQLite logging.

BikeShare Fleet Management System | Google Apps Script, Google Sheets & Forms

Oct 2025

- Cut tracking delays **80%** by architecting event-driven pipelines automating checkout, timers, and real-time state sync.
- Eliminated double-booking race conditions by engineering a centralized mutex lock across all form-triggered handlers.
- Achieved zero manual ops by engineering cron-triggered pipelines automating scheduling, detection, and batch alerting.

LEADERSHIP

Equity Leaders Program Tech Hub

2025 – Present

Founder & President

Remote(East Africa)

- Founded the ELP Tech Hub, the technology arm of the Equity Leaders Program (East Africa's premier scholarship program for top academic scholars), building a peer network spanning computer science, cybersecurity, and information systems.
- Organized **30+** technical webinars and coding workshops (REST APIs, DevSecOps, AI) with industry speakers, and led hands-on tutoring in programming fundamentals for new and current scholars.